STAT 201A Homework 2 Solutions

Theorem numbers refer to Casella & Berger (C&B). The tools used most often are

- **Definition 1.3.2 (conditional probability).** If $P(B) > 0$, then $P(A \mid B) = \dfrac{P(A \cap B)}{P(B)}$.
- **Law of total probability.** If $A_1, A_2, \ldots$ partition S, then $P(B) = \sum_i P(B \mid A_i) P(A_i)$. Combined with Definition 1.3.2, this gives **Bayes' rule** (Theorem 1.3.5).
- **Theorem 1.5.3.** F is a cdf if and only if (a) $\lim_{x \to -\infty} F(x) = 0$ and $\lim_{x \to \infty} F(x) = 1$; (b) F is nondecreasing; and (c) F is right-continuous.

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1. (Exercise 1.21) *A closet contains n pairs of shoes. If $2r$ shoes are chosen at random $(2r < n)$, what is the probability that there will be no matching pair in the sample?*

- **Sample space.** "At random" means every subset of $2r$ of the $2n$ shoes is equally likely, so there are $\binom{2n}{2r}$ outcomes.
- **Favorable outcomes.** No matching pair means the $2r$ shoes come from $2r$ **different** pairs.
  - Choose which $2r$ of the n pairs are used: $\binom{n}{2r}$ ways.
  - From each chosen pair, choose the left or the right shoe: 2^{2r} ways.

Hence

$$P(\text{no matching pair}) = \frac{\binom{n}{2r}\, 2^{2r}}{\binom{2n}{2r}}.$$

Check: with $n = 5$ and $r = 2$, the formula gives $\frac{5 \cdot 16}{210} = \frac{8}{21}$, which matches a direct enumeration of all $\binom{10}{4} = 210$ subsets.

Alternative (ordered draws). Draw the shoes one at a time. The first can be anything; the second must avoid the first one's mate; and so on:

$$\frac{2n}{2n} \cdot \frac{2n-2}{2n-1} \cdot \frac{2n-4}{2n-2} \cdots \frac{2n-2(2r-1)}{2n-2r+1},$$

which simplifies to the same answer.

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2. (Exercise 1.33) *5% of men and .25% of women are color-blind. A person chosen at random is color-blind. What is the probability that the person is male?*

Let M = "male" and C = "color-blind." We are given $P(M) = P(M^c) = \frac{1}{2}$, $P(C \mid M) = 0.05$, and $P(C \mid M^c) = 0.0025$.

- **Total probability:**

$$\begin{aligned} P(C) &= P(C \mid M)P(M) + P(C \mid M^c)P(M^c) \\ &= 0.05(0.5) + 0.0025(0.5) = 0.02625. \end{aligned}$$

- **Bayes' rule:**

$$P(M \mid C) = \frac{P(C \mid M)P(M)}{P(C)} = \frac{0.025}{0.02625} = \frac{20}{21} \approx 0.952.$$

Color blindness is 20 times as common among men, so about 20 of every 21 color-blind people are men.

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3. (Exercise 1.38) *Assume that any conditioning event has positive probability.*

(a) *If $P(B) = 1$, then $P(A \mid B) = P(A)$ for any A.*

- $P(B^c) = 1 - P(B) = 0$. Since $A \cap B^c \subset B^c$, monotonicity (Theorem 1.2.9(c)) gives $0 \le P(A \cap B^c) \le P(B^c) = 0$.
- A is the disjoint union of $A \cap B$ and $A \cap B^c$, so $P(A) = P(A \cap B) + P(A \cap B^c) = P(A \cap B)$.
- Therefore $P(A \mid B) = \dfrac{P(A \cap B)}{P(B)} = \dfrac{P(A)}{1} = P(A)$. $\square$

(b) *If $A \subset B$, then $P(B \mid A) = 1$ and $P(A \mid B) = P(A)/P(B)$.*

If $A \subset B$, then $A \cap B = A$. So

$$\begin{aligned} P(B \mid A) &= \frac{P(A \cap B)}{P(A)} = \frac{P(A)}{P(A)} = 1, \\ P(A \mid B) &= \frac{P(A \cap B)}{P(B)} = \frac{P(A)}{P(B)}. \quad \square \end{aligned}$$

(c) *If A and B are mutually exclusive, then $P(A \mid A \cup B) = \dfrac{P(A)}{P(A) + P(B)}$.*

- $A \subset A \cup B$, so $A \cap (A \cup B) = A$.
- $A \cap B = \emptyset$, so by Axiom 3, $P(A \cup B) = P(A) + P(B)$.

Therefore

$$P(A \mid A \cup B) = \frac{P\big(A \cap (A \cup B)\big)}{P(A \cup B)} = \frac{P(A)}{P(A) + P(B)}. \quad \square$$

(d) $P(A \cap B \cap C) = P(A \mid B \cap C)\, P(B \mid C)\, P(C)$.

By Definition 1.3.2, applied twice,

$$P(A \mid B \cap C)\,P(B \mid C)\,P(C) = \frac{P(A \cap B \cap C)}{P(B \cap C)} \cdot \frac{P(B \cap C)}{P(C)} \cdot P(C) \\ = P(A \cap B \cap C).$$

The assumption that conditioning events have positive probability ($P(B \cap C) > 0$, $P(C) > 0$) is what makes the cancellations legal. □

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4. (Exercise 1.39) *Suppose $P(A) > 0$ and $P(B) > 0$.* Then $P(A)P(B) > 0$.

(a) If A and B are mutually exclusive, then $A \cap B = \emptyset$, so $P(A \cap B) = 0 \neq P(A)P(B)$. Hence A and B are not independent. □

(b) If A and B are independent, then $P(A \cap B) = P(A)P(B) > 0$. Since $P(\emptyset) = 0$, the event $A \cap B$ cannot be empty, so A and B are not mutually exclusive. □

(Part (b) is the contrapositive of part (a).) The intuition: if A and B are mutually exclusive, learning that A occurred tells you for certain that B did not, which is as far from "no information" as possible.

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5. Two medical tests. We are given $P(D) = 0.01$, $P(T_i \mid D) = 0.95$, and $P(T_i \mid D^c) = 0.10$, with T_1 and T_2 independent given D and given D^c.

(a) By total probability,

$$P(T_1) = 0.95(0.01) + 0.10(0.99) = 0.1085,$$

and by Bayes' rule

$$P(D \mid T_1) = \frac{P(T_1 \mid D)P(D)}{P(T_1)} = \frac{0.0095}{0.1085} = \frac{19}{217} \approx 0.0876.$$

Even after a positive test, the chance of disease is under 9%, because healthy people are 99 times as common as sick ones.

(b) By conditional independence,

$$P(T_1 \cap T_2 \mid D) = 0.95^2 = 0.9025, \qquad P(T_1 \cap T_2 \mid D^c) = 0.10^2 = 0.01.$$

So by total probability,

$$P(T_1 \cap T_2) = 0.9025(0.01) + 0.01(0.99) = 0.018925,$$

and by Bayes' rule

$$P(D \mid T_1 \cap T_2) = \frac{0.009025}{0.018925} = \frac{361}{757} \approx 0.477.$$

(c) Use $p' = P(D \mid T_1) = \frac{19}{217}$ as the prior and apply Bayes' rule to test 2 alone:

$$\frac{0.95\,p'}{0.95\,p' + 0.10\,(1-p')} = \frac{0.95(19)}{0.95(19) + 0.10(198)} = \frac{18.05}{37.85} = \frac{361}{757},$$

the same answer as (b).

Why this works. Bayes' rule within the world where T_1 has occurred reads

$$P(D \mid T_1 \cap T_2) = \frac{P(T_2 \mid D \cap T_1)\,P(D \mid T_1)}{P(T_2 \mid D \cap T_1)\,P(D \mid T_1) + P(T_2 \mid D^c \cap T_1)\,P(D^c \mid T_1)}.$$

Conditional independence lets us drop T_1 from the likelihoods:

$$P(T_2 \mid D \cap T_1) = \frac{P(T_1 \cap T_2 \mid D)}{P(T_1 \mid D)} = \frac{P(T_1 \mid D)P(T_2 \mid D)}{P(T_1 \mid D)} = P(T_2 \mid D),$$

and likewise with D^c. Once the disease status is known, the first result says nothing more about the second, so updating one test at a time gives the same posterior as updating on both at once. Yesterday's posterior is today's prior.

(d)

$$P(T_1 \cap T_2) = 0.018925, \qquad P(T_1)P(T_2) = 0.1085^2 \approx 0.01177.$$

These differ, so T_1 and T_2 are **not** independent. In fact
$P(T_2 \mid T_1) = 0.018925/0.1085 \approx 0.174 > 0.1085 = P(T_2)$.

In words: both tests depend on the same hidden cause, the disease status. If we do not know that status, a positive first test makes disease more likely, which in turn makes a positive second test more likely. Conditioning on D (or D^c) fixes the common cause and removes the link, which is why the tests are independent given D but not unconditionally.

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6. (Exercise 1.47) We check the three conditions of Theorem 1.5.3, then classify X by Definition 1.5.7.

(a) $F(x) = \frac{1}{2} + \frac{1}{\pi}\tan^{-1}(x)$.

- **Limits:** $\tan^{-1}(x) \to \mp\frac{\pi}{2}$ as $x \to \mp\infty$, so $F(x) \to \frac{1}{2} - \frac{1}{2} = 0$ and $F(x) \to \frac{1}{2} + \frac{1}{2} = 1$.
- **Nondecreasing:** $F'(x) = \dfrac{1}{\pi(1+x^2)} > 0$.
- **Right-continuous:** F is continuous everywhere.

F is a continuous cdf, so X is **continuous**. (This is the Cauchy distribution, with pdf $F'(x)$.)

(b) $F(x) = e^{-e^{-x}}$.

- **Limits:** as $x \to -\infty$, $e^{-x} \to \infty$, so $F(x) \to e^{-\infty} = 0$. As $x \to \infty$, $e^{-x} \to 0$, so $F(x) \to e^0 = 1$.
- **Nondecreasing:** $F'(x) = e^{-x}e^{-e^{-x}} > 0$.
- **Right-continuous:** F is a composition of continuous functions, so it is continuous.

X is **continuous**, with pdf $f(x) = e^{-x}e^{-e^{-x}}$. (This is the Gumbel, or extreme value, distribution.)

(c) The function (1.5.6) is, for some $0 < \epsilon < 1$,

$$F_Y(y) = \begin{cases} \dfrac{1-\epsilon}{1+e^{-y}} & y < 0 \\[2ex] \epsilon + \dfrac{1-\epsilon}{1+e^{-y}} & y \geq 0. \end{cases}$$

- **Limits:** as $y \to -\infty$, $1 + e^{-y} \to \infty$, so $F_Y(y) \to 0$. As $y \to \infty$, $F_Y(y) \to \epsilon + (1-\epsilon) = 1$.
- **Nondecreasing:** $\frac{1}{1+e^{-y}}$ is increasing, so each piece is increasing. At the break, the left limit is $F_Y(0^-) = \frac{1-\epsilon}{2}$, and $F_Y(0) = \epsilon + \frac{1-\epsilon}{2} = \frac{1+\epsilon}{2}$, which is larger. So F_Y never decreases.
- **Right-continuous:** each piece is continuous, and $F_Y(0)$ is given by the right-hand piece, so $\lim_{y \downarrow 0} F_Y(y) = F_Y(0)$.

F_Y jumps by ϵ at 0, so it is **not continuous**. It increases strictly between jumps, so it is **not a step function**. Hence Y is **neither** continuous nor discrete: it has $P(Y = 0) = \epsilon$ plus a continuous part of total probability $1 - \epsilon$.

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7. A cdf with jumps.

(a)

- $\{X \leq b\}$ is the disjoint union of $\{X \leq a\}$ and $\{a < X \leq b\}$. So $F(b) = F(a) + P(a < X \leq b)$, that is, $P(a < X \leq b) = F(b) - F(a)$.
- Let $A_n = \{X \leq a - \frac{1}{n}\}$. Then $A_1 \subset A_2 \subset \cdots$, and $\bigcup_n A_n = \{X < a\}$: if $X(s) < a$, then $X(s) \leq a - \frac{1}{n}$ for all large n, and conversely every A_n lies inside $\{X < a\}$. By continuity from below,

  $$P(X < a) = \lim_{n\to\infty} P(A_n) = \lim_{n\to\infty} F\left(a - \frac{1}{n}\right) = F(a^-).$$

  The last step holds because F is nondecreasing and bounded, so its left limit exists and equals the limit along any sequence increasing to a.
- $\{X \leq a\}$ is the disjoint union of $\{X < a\}$ and $\{X = a\}$, so $P(X = a) = F(a) - F(a^-)$. $\square$

(b)

- **Limits:** $F(x) = 0$ for all $x < 0$, and $1 - \frac{1}{4}e^{-(x-2)} \to 1$ as $x \to \infty$.
- **Nondecreasing:** each piece is nondecreasing ($x/4$ increases, $\frac{1}{2}$ is flat, and $1 - \frac{1}{4}e^{-(x-2)}$ increases). At the breaks:
  - at 0: $F(0^-) = 0 = F(0)$;
  - at 1: $F(1^-) = \frac{1}{4} < \frac{1}{2} = F(1)$, a jump up of $\frac{1}{4}$;
  - at 2: $F(2^-) = \frac{1}{2} < \frac{3}{4} = F(2)$, a jump up of $\frac{1}{4}$.
- **Right-continuous:** each piece is continuous, and every interval in the definition is closed on the left, so $F(a)$ equals the right limit at each break.

F has jumps at 1 and 2, so it is not continuous. It increases on $(0, 1)$ and $(2, \infty)$, so it is not a step function. Hence X is **neither** continuous nor discrete.

(c) Using (a):

$$\begin{aligned} P(X = 1) &= F(1) - F(1^-) = \tfrac{1}{2} - \tfrac{1}{4} = \tfrac{1}{4}, \\ P(X = 2) &= F(2) - F(2^-) = \tfrac{3}{4} - \tfrac{1}{2} = \tfrac{1}{4}, \\ P\left(\tfrac{1}{2} < X < 2\right) &= F(2^-) - F\left(\tfrac{1}{2}\right) = \tfrac{1}{2} - \tfrac{1}{8} = \tfrac{3}{8}, \\ P(X > 3) &= 1 - F(3) = \tfrac{1}{4}e^{-1} \approx 0.092. \end{aligned}$$

For the conditional probability, $\{X \geq 1\} \cap \{X \leq 2\} = \{1 \leq X \leq 2\}$, so

$$P(X \geq 1 \mid X \leq 2) = \frac{F(2) - F(1^-)}{F(2)} = \frac{\frac{3}{4} - \frac{1}{4}}{\frac{3}{4}} = \frac{2}{3}.$$

Watch the endpoints: $P(\frac{1}{2} < X < 2)$ excludes the jump at 2, and $P(X \geq 1)$ includes the jump at 1.

(d) The jumps carry total mass $\frac{1}{4} + \frac{1}{4} = \frac{1}{2}$, and the increasing parts carry the other $\frac{1}{2}$: slope $\frac{1}{4}$ on $(0, 1)$ gives $\frac{1}{4}$, and density $\frac{1}{4}e^{-(x-2)}$ on $(2, \infty)$ gives $\frac{1}{4}$. So the coin is fair. Flip a fair coin:

- **Heads:** draw X from the discrete distribution $P(X = 1) = P(X = 2) = \frac{1}{2}$.
- **Tails:** draw X from the continuous density

$$g(x) = \begin{cases} \frac{1}{2} & 0 < x < 1 \\ \frac{1}{2}e^{-(x-2)} & x > 2 \\ 0 & \text{otherwise,} \end{cases}$$

  which integrates to $\frac{1}{2} + \frac{1}{2} = 1$.

Check that $F = \frac{1}{2}F_{\text{disc}} + \frac{1}{2}F_{\text{cont}}$ (total probability over the coin):

- $0 \leq x < 1$: $\frac{1}{2}(0) + \frac{1}{2} \cdot \frac{x}{2} = \frac{x}{4}$;
- $1 \leq x < 2$: $\frac{1}{2} \cdot \frac{1}{2} + \frac{1}{2} \cdot \frac{1}{2} = \frac{1}{2}$;
- $x \geq 2$: $\frac{1}{2}(1) + \frac{1}{2}\left[\frac{1}{2} + \frac{1}{2}\left(1 - e^{-(x-2)}\right)\right] = 1 - \frac{1}{4}e^{-(x-2)}$.

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8. (Exercise 1.55) T is exponential with mean 1.5, so $P(T \le t) = 1 - e^{-t/1.5}$ for $t > 0$.

$V = 5$ if $T < 3$, and $V = 2T \ge 6$ if $T \ge 3$. So V takes the value 5 or a value in $[6, \infty)$.

- $v < 5$**:** V is never below 5, so $F_V(v) = 0$.
- $5 \le v < 6$**:** $V \le v$ happens only when $V = 5$, that is, when $T < 3$:

$$F_V(v) = P(T < 3) = 1 - e^{-3/1.5} = 1 - e^{-2}.$$

- $v \ge 6$**:** $V \le v$ happens when $T < 3$, or when $3 \le T$ and $2T \le v$. These are disjoint and together make up $\{T \le v/2\}$:

$$F_V(v) = P(T < 3) + P\left(3 \le T \le \frac{v}{2}\right) = P\left(T \le \frac{v}{2}\right) = 1 - e^{-v/3}.$$

Hence

$$F_V(v) = \begin{cases} 0 & v < 5 \\ 1 - e^{-2} & 5 \le v < 6 \\ 1 - e^{-v/3} & v \ge 6. \end{cases}$$

- F_V jumps by $1 - e^{-2} \approx 0.865$ at $v = 5$, which is $P(V = 5)$.
- F_V is flat on $[5, 6)$, since V never lands there, and it is **continuous** at $v = 6$: $1 - e^{-6/3} = 1 - e^{-2}$.
- Like Problem 7, V is neither discrete nor continuous.

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9. A pmf and a pdf with the same tail.

(a) The partial sums telescope:

$$\sum_{k=1}^{N} \frac{c}{k(k+1)} = c \sum_{k=1}^{N} \left(\frac{1}{k} - \frac{1}{k+1}\right) = c\left(1 - \frac{1}{N+1}\right) \xrightarrow{N \to \infty} c.$$

A pmf must sum to 1, so $c = 1$. With $c = 1$, $f_X(k) > 0$ for every k, so nonnegativity also holds.

(b) For $x < 1$, $F_X(x) = 0$. For $k \le x < k + 1$ with $k \ge 1$, the same telescoping gives $F_X(x) = \sum_{j=1}^{k} \frac{1}{j(j+1)} = 1 - \frac{1}{k+1}$. So

$$F_X(x) = \begin{cases} 0 & x < 1 \\ 1 - \dfrac{1}{\lfloor x \rfloor + 1} & x \ge 1. \end{cases}$$

- **Sketch:** a right-continuous step function, equal to $\frac{1}{2}$ on $[1, 2)$, $\frac{2}{3}$ on $[2, 3)$, $\frac{3}{4}$ on $[3, 4)$, and so on, rising toward 1. The jump at each integer $k \ge 1$ has height $\frac{1}{k(k+1)}$. Each step includes its left endpoint (solid dot) and excludes its right endpoint (open dot).

- **Tail:** $P(X > n) = 1 - F_X(n) = \dfrac{1}{n+1}$.

(c) Normalization:

$$\int_1^\infty \frac{c}{y^2}\,dy = c\left[-\frac{1}{y}\right]_1^\infty = c, \quad \text{so } c = 1.$$

Then $F_Y(y) = 0$ for $y \le 1$, and for $y > 1$,

$$F_Y(y) = \int_1^y \frac{1}{t^2}\,dt = 1 - \frac{1}{y}.$$

Since Y is continuous, $P(Y = n) = 0$, so for $n \ge 1$, $P(Y \ge n) = 1 - F_Y(n) = \dfrac{1}{n}$.

(d) $Y > 1$, so $\lfloor Y \rfloor$ takes the values $1, 2, \ldots$ For $k \ge 1$, $\{\lfloor Y \rfloor = k\} = \{k \le Y < k+1\}$, so

$$P(\lfloor Y \rfloor = k) = F_Y(k+1) - F_Y(k) = \frac{1}{k} - \frac{1}{k+1} = \frac{1}{k(k+1)} = f_X(k).$$

$\lfloor Y \rfloor$ and X have the same support and the same probabilities, so they have the same distribution. $\square$

(e)

- **A pmf is a probability.** $f_X(x) = P(X = x)$, and $\{X = x\} \subset S$, so $f_X(x) \le P(S) = 1$ by monotonicity.
- **A pdf need not be ≤ 1.** A density value is not a probability; only its integral over a set is. For example, the uniform$(0, \frac{1}{2})$ pdf

  $$f(y) = \begin{cases} 2 & 0 < y < \frac{1}{2} \\ 0 & \text{otherwise} \end{cases}$$

  is nonnegative and integrates to $2 \cdot \frac{1}{2} = 1$, yet equals 2 everywhere on its support.
- A pdf can even be **unbounded**: $f(y) = \frac{1}{2\sqrt{y}}$ on $(0, 1)$ integrates to $\sqrt{y}\,\big|_0^1 = 1$ but $f(y) \to \infty$ as $y \downarrow 0$.